
Pin Map

Pin Name	Pin	Pin Name	Pin
SH/LD	1	CLK	2
E	3	F	4
G	5	H	6
QHnot	7	QH	13
SER	14	A	15
B	16	C	17
D	18	CLK INH	19

Global Parameters

Parameter	Value
VCC Pin	20
GND Pin	8
VCC Voltage	2.5, 3.3, 4.0, 4.5, 5
Input Low	0.75, 0.99, 1.2, 1.35, 1.5
Input High	1.75, 2.31, 2.8, 3.15, 3.5
Output Low	0.3, 0.3, 0.3, 0.3, 0.3
Output High	2.2, 3.0, 3.6, 4.1, 4.7

Tests: **IC FAIL 0/5**

Parallel Load: **FAIL**

Inputs				VCC	Outputs/Results		
SH/LD	CLK INH	A, B, C, D, E, F, G, H	CLK		QH	QHnot	
L	X	H, L, H, L, H, L, H, H	X	2.5	2.37 (H)	X	0.0 (L)
				3.3	3.15 (H)		0.0 (L)
				4.0	3.8 (H)		0.0 (L)
				4.5	4.39 (H)		0.0 (L)
				5	4.87 (H)		0.0 (L)
H	L	H, L, H, L, H, L, H, H	R	2.5	2.37 (H)	L	0.0 (L)
				3.3	3.15 (H)		0.0 (L)
				4.0	3.8 (H)		0.0 (L)
				4.5	4.39 (H)		0.0 (L)
				5	4.87 (H)		0.0 (L)
H	L	H, L, H, L, H, L, H, H	R	2.5	0.0 (L)	L	2.37 (H)
				3.3	0.0 (L)		3.15 (H)
				4.0	0.0 (L)		3.81 (H)
				4.5	0.0 (L)		4.43 (H)
				5	0.0 (L)		4.87 (H)
H	L	H, L, H, L, H, L, H, H	R	2.5	2.37 (H)	H	0.0 (L)
				3.3	3.15 (H)		0.0 (L)
				4.0	3.8 (H)		0.0 (L)
				4.5	4.39 (H)		0.0 (L)
				5	4.87 (H)		0.0 (L)

Inputs				VCC	Outputs/Results			
SH/LD	CLK INH	A, B, C, D, E, F, G, H	CLK		QH		QHnot	
H	L	H, L, H, L, H, L, H, H	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.43 (H)
				5		0.0 (L)		4.87 (H)
H	L	H, L, H, L, H, L, H, H	R	2.5	L	2.37 (H)	H	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.8 (H)		0.0 (L)
				4.5		4.39 (H)		0.0 (L)
				5		4.87 (H)		0.0 (L)
H	L	H, L, H, L, H, L, H, H	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.43 (H)
				5		0.0 (L)		4.87 (H)
H	L	H, L, H, L, H, L, H, H	R	2.5	L	2.37 (H)	H	0.0 (L)
				3.3		3.15 (H)		0.0 (L)
				4.0		3.8 (H)		0.0 (L)
				4.5		4.39 (H)		0.0 (L)
				5		4.87 (H)		0.0 (L)

Inputs				VCC	Outputs/Results		
SH/LD	CLK INH	A, B, C, D, E, F, G, H	CLK		QH		QHnot
H	L	H, L, H, L, H, L, H, H	R	2.5	H	2.37 (H)	0.0 (L)
				3.3		3.15 (H)	0.0 (L)
				4.0		3.8 (H)	0.0 (L)
				4.5		4.4 (H)	0.0 (L)
				5		4.87 (H)	0.0 (L)

Serial In 1: FAIL

Inputs				VCC	Outputs/Results		
SH/LD	SER	CLK	CLK INH		QH		QHnot
H	H	L	R	2.5	X	0.0 (L)	2.37 (H)
				3.3		0.0 (L)	3.15 (H)
				4.0		0.0 (L)	3.81 (H)
				4.5		0.0 (L)	4.45 (H)
				5		0.0 (L)	4.87 (H)
H	L	L	R	2.5	X	0.0 (L)	2.37 (H)
				3.3		0.0 (L)	3.15 (H)
				4.0		0.0 (L)	3.81 (H)
				4.5		0.0 (L)	4.44 (H)
				5		0.0 (L)	4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK	CLK INH		QH		QHnot	
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK	CLK INH		QH		QHnot	
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK	CLK INH		QH		QHnot	
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK	CLK INH		QH		QHnot	
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Serial In 2: FAIL

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	L	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	L	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.45 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.45 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	X	0.0 (L)	X	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	H	L	R	2.5	H	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.45 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	H	L	R	2.5	L	0.0 (L)	H	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	SER	CLK INH	CLK		QH		QHnot	
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

No Change 1: FAIL

Inputs				VCC	Outputs/Results			
SH/LD	H	CLK INH	CLK		QH		QHnot	
L	H	X	X	2.5	X	2.37 (H)	X	0.0 (L)
				3.3		3.14 (H)		0.0 (L)
				4.0		3.49 (U)		0.0 (L)
				4.5		4.11 (H)		0.02 (L)
				5		4.86 (H)		0.0 (L)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	H	CLK INH	CLK		QH		QHnot	
H	L	X	H	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	X	H	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	X	H	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

No Change 2: **FAIL**

Inputs				VCC	Outputs/Results			
SH/LD	H	CLK INH	CLK		QH		QHnot	
L	H	X	X	2.5	X	2.37 (H)	X	0.0 (L)
				3.3		3.14 (H)		0.0 (L)
				4.0		3.76 (H)		0.0 (L)
				4.5		4.16 (H)		0.02 (L)
				5		4.86 (H)		0.0 (L)
H	H	L	R	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	H	X	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)
H	L	H	X	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)

Inputs				VCC	Outputs/Results			
SH/LD	H	CLK INH	CLK		QH		QHnot	
H	L	H	X	2.5	H	0.0 (L)	L	2.37 (H)
				3.3		0.0 (L)		3.15 (H)
				4.0		0.0 (L)		3.81 (H)
				4.5		0.0 (L)		4.44 (H)
				5		0.0 (L)		4.87 (H)